01 / APPLICATION SECURITY

SQL Injection: Attack, Fix, Verify

Node.js | Express | SQLite | bcrypt | HTTP tests

A controlled local web security lab showing how an unsafe login query can allow authentication bypass, how a parameterized query prevents it, and how tests verify the difference.

THE PROBLEM AND THE FIX

VULNERABLE ROUTE

User input is inserted directly into the SQL query string.

```
' OR '1'='1
```

Result: login bypass in the lab

SECURE ROUTE

The username is bound as data; bcrypt verifies the password hash.

```
WHERE username = ?
```

Result: payload rejected

WHAT I BUILT AND VERIFIED

1. Kept vulnerable and secure login routes side by side for a reproducible comparison.
2. Documented root cause, impact, remediation, and limits in FINDINGS.md.
3. Wrote 3 HTTP tests: bypass succeeds in the teaching route; the secure route rejects it; a normal login succeeds.

HOW TO DEMONSTRATE IT AT THE FAIR

1. Enter the demonstration payload in both fields of the vulnerable local login.
2. Repeat it on the secure login and show that access is denied.
3. Run npm test, then explain the parameterized query and the test results.

PROJECT SOURCE AND CASE STUDY

https://github.com/atulyado/SQL-Injection-Attack-Fix-Verify

02 / NETWORK SECURITY

Packet Sniffer

Python | Scapy | TCP / UDP / ICMP | CLI

A command-line project for observing packet traffic and learning how network protocols appear in a capture. It records packet details and raises a simple high-volume alert.

CAPTURE AND ANALYSIS WORKFLOW

CAPTURE

Scapy reads packets

PARSE

Extract addresses, protocol, time

FILTER & LOG

Show selected traffic, append logs

ALERT

Flag high packet rate from one IP

IMPLEMENTED FEATURES

- Displays source and destination IP addresses, protocol, packet count, and timestamp.
- Supports TCP, UDP, and ICMP filters plus a selected packet-count limit.
- Writes packet activity to a log file for later review.
- Uses a simple per-source traffic threshold to flag potentially unusual bursts.

WHAT THE ALERT MEANS

The threshold is an educational heuristic. A busy host can trigger it without being malicious, so the alert is a starting point for investigation rather than proof of an attack.

PROJECT SOURCE AND RUN INSTRUCTIONS

https://github.com/atulyado/Packet-Sniffer